

Deep learning phase pickers: how well can existing models detect hydraulic-fracturing induced microseismicity from a borehole array?

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SUMMARY

Deep learning (DL) phase picking models have proven effective in processing large volumes of seismic data, including successfully detecting earthquakes missed by other standard detection methods. Despite their success, the applicability of existing extensively trained DL models to high-frequency borehole data sets is currently unclear. In this study, we compare four established models [Generalized Seismic Phase Detection (GPD), U-GPD, PhaseNet and EQTransformer] trained on regional earthquakes recorded at surface stations (100 Hz) in terms of their picking performance on high-frequency borehole data (2000 Hz) from the Preston New Road (PNR) unconventional shale gas site, in the United Kingdom (UK). The PNR-1z data set, which we use as a benchmark, consists of continuously recorded waveforms containing over 38 000 seismic events previously catalogued, ranging in magnitudes from -2.8 to 1.1 . Remarkably, all four DL models can detect induced seismicity in high-frequency borehole data and two might satisfy the monitoring requirements of some users without any modifications. In particular, PhaseNet and U-GPD demonstrate exceptional recall rates of 95 and 76.6 per cent, respectively, and detect a substantial number of new events (over 15 800 and 8300 events, respectively). PhaseNet's success might be attributed to its exposure to more extensive and diverse instrument data set during training, as well as its relatively small model size, which might mitigate overfitting to its training set. U-GPD outperforms PhaseNet during periods of high seismic rates due to its smaller window size (400 samples compared to PhaseNet's 3000-sample window). These models start missing events below $M_w -0.5$, suggesting that the models could benefit from additional training with microseismic data-sets. Nonetheless, PhaseNet may satisfy some users' monitoring requirements without further modification, detecting over 52 000 events at PNR. This suggests that DL models can provide efficient solutions to the big data challenge of downhole monitoring of hydraulic-fracturing induced seismicity as well as improved risk mitigation strategies at unconventional exploration sites.

Key words: Machine learning; Neural networks, fuzzy logic; Downhole methods; Computational seismology; Earthquake monitoring and test-ban treaty verification; Induced seismicity.

1 INTRODUCTION

Hydraulic-fracturing induced seismicity (HFIS) can pose serious risks to a country's infrastructure, energy security and communities (Li *et al.* 2019; Atkinson *et al.* 2020; Schultz *et al.* 2020). Examples of significant HFIS include the 2011 moment magnitude M_w 4.8 Eagle Ford earthquake in Texas, United States (Frohlich & Brunt 2013), the 2016 M_w 4.1 Fox Creek earthquake in Alberta, Canada (Schultz *et al.* 2017) and the 2017 M_w 4.7 Changning earthquake in South Sichuan, China (Lei *et al.* 2017). The risks extend to wastewater disposal sites (Chen *et al.* 2017), enhanced geothermal

sites (Grigoli *et al.* 2018) and potentially carbon capture and storage sites (Verdon & Stork 2016).

Microseismic monitoring of HFIS by operators is critical for potential risk mitigation measures and is a legal requirement in many countries (e.g. Wong *et al.* 2015; Kao *et al.* 2016; Clarke *et al.* 2019). In addition, high-resolution data sets of small events can lead to a better understanding, modelling and forecasting of the mechanisms and hazards of HFIS (e.g. Eyre *et al.* 2019; Kettlety & Verdon 2021; Mancini *et al.* 2022). To that end, arrays of high-frequency geophones installed in boreholes close to stimulation wells are particularly useful for detecting and characterizing the

quickly attenuating high-frequency content of small events (Klinger & Werner 2022; Holmgren *et al.* 2023).

However, borehole arrays with high sampling frequencies can generate substantial volumes of data, as monitoring arrays may include tens to hundreds of seismic stations. Examples of large borehole data sets include the Tony Creek Dual Microseismic Experiment (ToC2ME) data set in Fox Creek, Alberta (Eaton *et al.* 2018), the Horn River basin data set in British Columbia, Canada (Verdon & Budge 2018) and the Frontier Observatory for Research in Geothermal Energy (FORGE) data set in Utah, United States (Shi *et al.* 2022). Processing large data sets requires significant time and costly computational resources, posing an issue in generating event catalogues that are crucial for informing our understanding of subsurface activities, especially in real time.

Deep learning (DL) phase pickers offer a solution to efficiently picking seismic events in large volumes of continuous seismic data. Perol *et al.* (2018), Ross *et al.* (2018), Zhu & Beroza (2019) and Mousavi *et al.* (2020) demonstrated that DL models can pick events more efficiently than standard approaches and human analysts. Specifically, DL neural networks such as Generalized Seismic Phase Detection (GPD, Ross *et al.* 2018), U-GPD (Lapins *et al.* 2021), EQTransformer (Mousavi *et al.* 2020) and PhaseNet (Zhu & Beroza 2019) have undergone extensive training to proficiently pick seismic phases within large data sets. Moreover, DL phase pickers have detected new events overlooked by conventional methods, thereby contributing, for instance, to uncovering the complexities of fault structures (Tan *et al.* 2021) and earthquake swarm dynamics, providing new insights into aseismic crustal processes (Ross & Cochran 2021).

The aim of this paper is to assess the performances of various existing DL models when picking microseismic phases in high-frequency borehole data without training. Most existing DL models, including GPD, U-GPD, EQTransformer and PhaseNet, are trained using 100 Hz data collected from surface seismic stations. These training sets mainly consist of larger, regional-sized earthquakes ($M_w > 0$) but also encompass a number of smaller, local earthquakes ($M_w < 0$). However, it is not clear that the models can generalize to detect microseismicity in high-frequency data in a borehole setting. In this study, we apply and compare these models using the Preston New Road 1z (PNR-1z) shale gas exploration data set (Clarke *et al.* 2019), where the distance between the centroids of the event hypocentres and the borehole array is relatively small (328.7 m). This benchmark catalogue contained over 38 000 events, with moment magnitudes M_w ranging from -2.8 to 1.1 , recorded on an array of 24 borehole geophones during injection activities at a sampling rate of 2000 Hz (Fig. 1). Furthermore, we assess whether DL models identify additional microseismic events, and we compare each model in terms of phase detection and picking ability.

2 DL MODELS

We evaluate four DL models (Table 1): the GPD model by Ross *et al.* (2018), the U-GPD model by Lapins *et al.* (2021), EQTransformer (EQT) by Mousavi *et al.* (2020) and PhaseNet by Zhu & Beroza (2019). We select these models because they have publicly available code, working GitHub repositories, and the models were extensively trained on large data sets for generalized phase picking. Additionally, a previous study by Münchmeyer *et al.* (2022) found

that GPD, PhaseNet and EQT were the best-performing models for earthquake detection, phase classification and onset time determination for regional and teleseismic data sets (not including downhole, high-frequency data sets). These models, however, were primarily trained on 100 Hz three-component seismograms from surface seismic stations and were not specifically trained, nor tested, on high-frequency borehole data sets.

GPD is a convolutional neural network (CNN) trained on 4.5 million waveforms (1.5 million each for P , S and noise). Ross *et al.* (2018) trained this model with magnitudes M from -0.81 to 5.7 recorded by the Southern California Seismic Network. The model uses a 400-sample sliding window on continuous three-component data to input into convolutional layers for feature extraction. These extracted features are then input into a fully connected neural network (FCNN) for phase classification. The model outputs single class probability values for P , S and noise (i.e., model output dimensions are 3×1) for each window. When the phase probability is above a user-defined threshold, a phase is declared in the middle of the window.

U-GPD is a DL model that modifies the GPD model by using a fully convolutional U-Net architecture and fine-tuning the base GPD weights with an additional data set. Lapins *et al.* (2021) fine-tuned the weights using a limited volcano-seismic data set (from the Nabro volcano, Eritrea) sampled at 100 Hz. The Nabro data set contains 2498 event waveforms with local magnitudes ranging from -0.4 to 3.6 . U-GPD uses the same input as GPD (3×400 sliding window) but replaces the FCNN with additional convolutional layers. The new layers were initialized with randomized weights and then trained on the Nabro data set. As U-GPD is a fully convolutional model, its output differs from GPD as U-GPD estimates a class probability for each sample in the window (i.e. model output dimensions are 3×400). The probability traces help the network pick more precisely as it removes the ambiguity arising from a single class prediction over an entire signal window (Lapins *et al.* 2021).

The PhaseNet model, similar to U-GPD, is a U-Net fully CNN developed by Zhu & Beroza (2019). However, unlike GPD and U-GPD, PhaseNet uses a larger 3000-sample sliding window as its input. PhaseNet's training data set consists of over 600 000 event waveforms from the Northern California Earthquake Data Center. Their training data set comprises a magnitude range from M 0 to 5. A unique characteristic of this training data set is that Zhu & Beroza (2019) included different types of instruments (e.g. accelerometers, high gain and low gain seismometers) during training, which might help generalize its ability to phase pick in different data. The only modification we apply to PhaseNet is the change in the 'sampling_rate' variable within the 'data_reader.py' file from 100 to 2000 Hz so that it reads our input data correctly.

EQT is an attention-based DL model trained using 1 million labelled earthquake waveforms and 300 000 noise waveforms Mousavi *et al.* (2020) from the STanford EArthquake Dataset (STEAD), a global data set of earthquakes. Out of the models in this study, EQT has the largest data input window of 6000 samples. The model has an encoder, which extracts high level representations of the data from the continuous seismic signals and three decoder branches that use these data representations to generate three probability traces (for the presence of an earthquake, the P and S waves, respectively). EQT consists of convolutional layers (CNN), long-short-term-memory (LSTM) layers and a hierarchical attention mechanism. EQT's attention mechanism visually weights

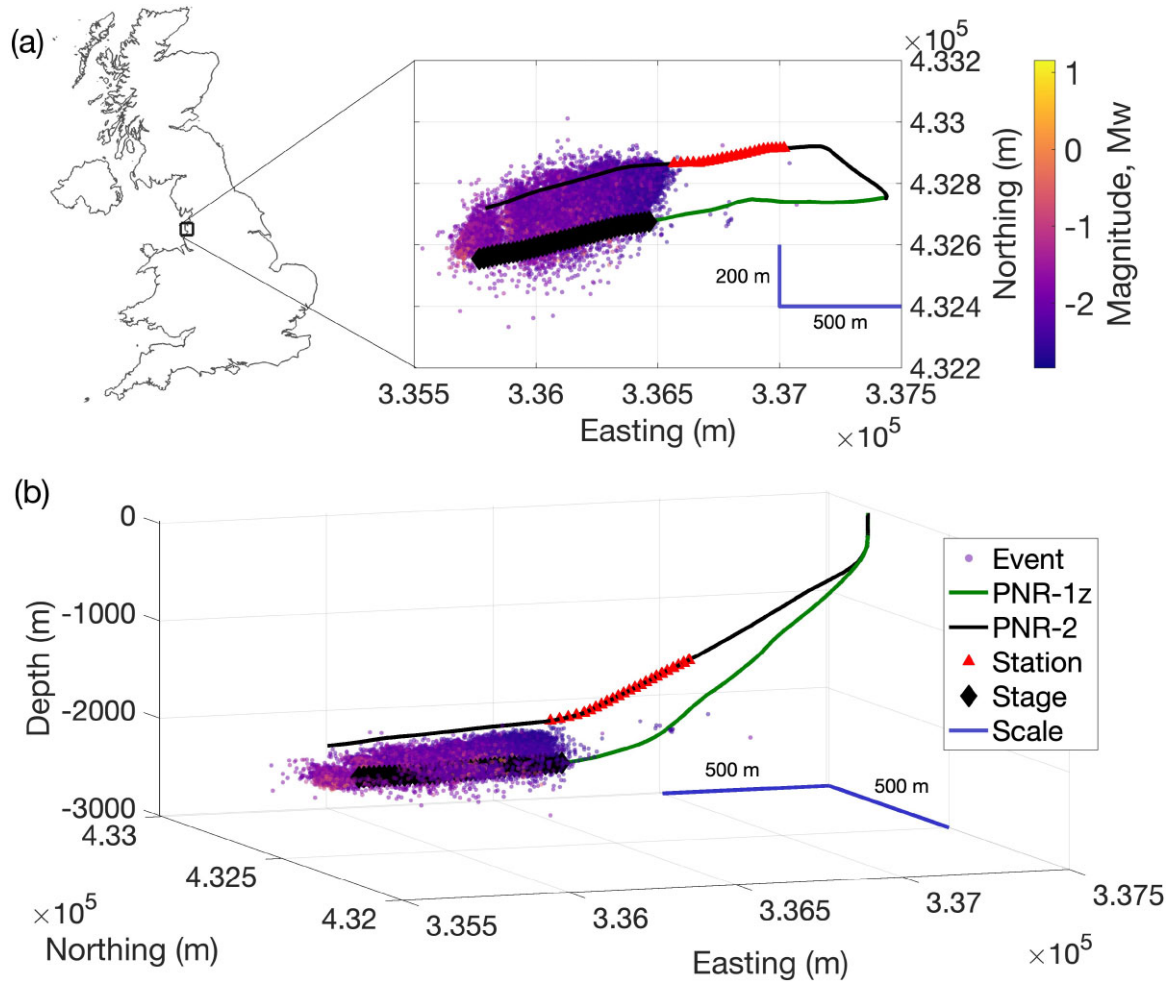


Figure 1. A geographical map showing the (a) plan view of the PNR unconventional shale gas exploration site and a (b) 3-D section of the site including seismicity (circles), geophones (triangles), stages (diamonds) on the PNR-1z and PNR-2 wells.

Table 1. The DL phase pickers used in this study. We include the input window sizes in samples, the input in time for 100 and 2000 Hz data in seconds, the magnitude ranges for the training data set, the architecture types of the models and the number of trainable parameters for each model.

Models	GPD	U-GPD	EQTransformer	PhaseNet
Input window size	400	400	6000	3000
Input (100 Hz)	4 s	4 s	60 s	30 s
Input (2000 Hz)	0.2 s	0.2 s	3 s	1.5 s
Magnitude range	−0.81 to 5.7	−0.81 to 5.7 and −0.4 to 3.6	−0.5 to 7.9	0 to 5
Architecture type	CNN + FCNN	U-Net	CNN + LSTM + attention	U-Net
No. of params	1,741,003	672,419	376,935	269,675

sections of data in its 6000-sample input window both globally (full waveform) and locally (where it expects the *P* and *S* phases). Earthquake magnitudes in STEAD range from *M* −0.5 to 7.9. EQ-Transformer comes in two versions: the ‘original’ model and the ‘conservative’ model. Both models are trained using the same training data sets, but their hyperparameters are optimized for different levels of event detection. The ‘original’ model is tuned to maximize the number of detected events whereas the ‘conservative’ version is tuned to minimize the number of false positives. For our study, we use the ‘original’ EQT version that has been optimized to maximize the number of detections. As the ‘original’ model can produce a large number of declarations that are not seismic events (Scotto di Uccio *et al.* 2023; Yoon *et al.* 2023), we further validate these

declarations in Section 4.4. The EQT model from the repository automatically resamples data to 100 Hz, automatically filters the data from 1 to 45 Hz and its sliding window is shifted in time (overlap $\times$ 60 s) which is suited for 100 Hz. We update this code so that the model reads our input data correctly by not resampling to 100 Hz, removing the in-built filter and correcting the time-shift window for 2000 Hz data (overlap $\times$ 3 s).

Applying these diverse DL models allows for a comprehensive comparison considering variations in model architecture, input window sizes, training data and the range of trained earthquake magnitudes (Table 1). This assessment aims to identify the model specifications that are most advantageous for monitoring microseismicity in high-frequency borehole data.

3 DATA

3.1 Preston New Road continuous downhole data set

Cuadrilla Resources Ltd. monitored the hydraulic stimulation of the PNR-1z well using a borehole geophone array in the PNR-2 well (Clarke *et al.* 2019). Fig. 1 shows the downhole array of 24 Avalon Geochain Slimline 15 Hz geophones that continuously recorded seismic data of the operations that took place from 2018 October 15 to December 17. Hydraulic fracturing operations were carried out from October 15 to November 2, followed by an injection hiatus from 2018 November 3 to December 7 and then further stimulations from 2018 December 8 to 17. The distance between geophones in the borehole array is 30 m. The average distance between the centroids of the geophones and the injection stages is approximately 878 m.

The PNR-1z continuous downhole data set was recorded at 2000 Hz from 2018 October 8 to December 18 and a total data volume of 3.92 Tb was acquired, available in 16-s SEG-Y files. Each SEG-Y file contains three-component seismic data for 24 stations, resulting in 72 traces per file. These continuous traces serve as input to the DL models.

From visual inspection of the waveforms, we observe that the amplitudes of events above moment magnitude $M_w -0.237$ are clipped because the operators set a high gain on the recording instruments to detect more microseismic events (Verdon, personal communication). In previous work, Lim (2021) visually checked and found that seismograms of earthquakes down to $M_w -0.237$ are clipped on at least one station. Assuming all events down to $M_w -0.237$ are clipped, approximately 133 of the 38 452 previously catalogued events (0.3 per cent) are clipped. Although 133 of the larger events were clipped, this should not greatly affect phase detection as DL models that employ relatively short sliding windows will primarily focus on the onset of phase arrivals (Ross *et al.* 2018). Zhu & Beroza (2019) have also shown that PhaseNet is successfully able to pick phases even in strongly clipped data. Still, we consider the effects of clipping in Section 5.3, for these models and more complex phase picking models that exploit the full waveform content with longer time windows (i.e. EQTransformer).

3.2 Existing catalogue and injection data

We use the Coalescence Microseismic Mapping (CMM) catalogue as a benchmark to compare with the other DL models. Cuadrilla's contracted processor produced a seismic catalogue of event origin times, locations and magnitudes, from the PNR-1z data set, using proprietary code based on the CMM method (Drew *et al.* 2013). The CMM catalogue contains event origin times and locations of a total of 38 452 events with magnitudes that span $-2.839 \leq M_w \leq 1.155$. Fig. 2 shows the events and their magnitudes overlaid with the injection rate data over time.

The CMM method is a computationally intensive multistation simultaneous detection and location approach that generates characteristic functions for each station using short-term average over long-term average (STA/LTA) ratios (Drew *et al.* 2013). The migration of signals from multiple stations to a coherent source in time and space makes CMM a robust method for earthquake detection, with the added advantage of simultaneously determining hypocentres. While it has been proven to be a robust method for microseismic monitoring (Smith *et al.* 2015), it is constrained by the sliding time windows it employs on continuous data, detecting only the largest event (energy maxima) within a fixed time window. We set the CMM as a good benchmark for these single station DL

models while also exploring whether they can outperform the CMM method and provide a more efficient alternative for monitoring microseismicity.

4 METHODS

4.1 Data pre-processing

We pre-process the raw continuous data by rotating the waveforms from their respective station orientations to the E, N and Z components. For each model's input, we then apply a 50 Hz Butterworth high-pass filter. This ensures that the pre-trained models can pick microseismic phases within the noisy downhole data as microseismicity typically has low signal-to-noise ratio (SNR). We choose a 50 Hz high-pass filter by examining the fast Fourier transform event spectra on the geophones closest to (deepest) and farthest from (most shallow) the source (perforations in the PNR-1z well). Holmgren *et al.* (2021) also highlight electrical noise occurring below 50 Hz in the PNR-1z data. In addition, as most of the events are smaller than $M_w 1$, their corner frequencies would likely be above 50 Hz (Shearer 2019) and so, the 50 Hz high-pass filter is suitable for event detection.

4.2 Classification test

We first compare each model's ability to classify seismic phases (P , S and noise) in the PNR-1z continuous downhole waveform data from single stations, which we refer to as phase classification tests. Our test data set comprises 750 sections of data from 250 random events (i.e., 250 P , 250 S and 250 noise) on random stations. We then record the class labels produced by each model for each section. Each section is a 3-s (6000 sample) window of three-component continuous data containing a phase arrival in the window. We filter, visually inspect, and manually select each section from the CMM catalogue. We also randomly select stations to avoid bias, especially as some stations are less noisy or closer to events than other stations (e.g. shallow stations are further away from the events). If a model produces more than one label in the section, we record the class label closest to the manually determined pick time.

We calculate recall, precision and F1-scores for each phase as well as overall metrics to assess the classification performance of each model. For overall recall and precision, we compute the average values for each class (P , S and noise). Overall F1-score is calculated by using the average precision and recall values.

Recall measures the completeness of positive predictions and is calculated as

$$\text{Recall} = \frac{(\text{TP})}{(\text{TP} + \text{FN})}, \quad (1)$$

where TP is the number of true positives and FN is the number of false negatives.

For example, when calculating the recall value for the ' P ' phase of a model, we determine that: true positives (TP) are the number of ' P ' labels correctly identified as ' P '; false negatives (FN) are the number of ' P ' labels that the model misclassified as ' S ' or 'noise'; and their sum (TP + FN) is the total number of actual ' P ' labels in the test data set, which is fixed at 250.

Recall values close to 1 signify that the model produces a low number of false negatives. For example, a high R value for the P phase shows that the model does not miss a lot of P labels that are in the test set.

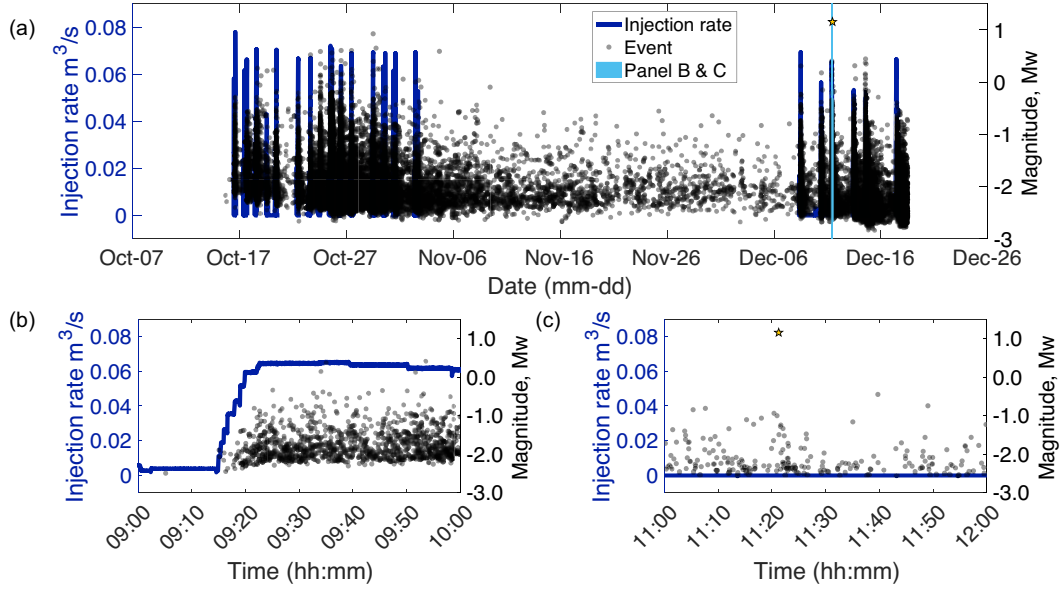


Figure 2. Temporal plot of the injection rates (blue line, left-hand axis) and CMM catalogued events (circles) with their respective moment magnitudes, M_w , (right-hand axis) for (a) the whole duration of the PNR-1z continuous downhole data, (b) during a period of high injection rates (2018 December 11, 9am–10am) and (c) during a period without injection (2018 December 11, 11am–12pm). The largest earthquake (M_w 1.1) of the catalogue is shown as a star.

Precision measures the accuracy of positive predictions by the model and is calculated as

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (2)$$

where FP is the number of false positives. When calculating the precision value for the ‘P’ phase of a model, we define: TP are the number of ‘P’ labels correctly identified as ‘P’; FP are the number of ‘S’ or ‘noise’ labels that the model misclassified as ‘P’; their sum (TP + FP) is the total number of ‘P’ classifications that the model predicted in the test data set.

High precision (close to 1) signifies a low number of false positive classifications (i.e. a high precision shows that the model does not falsely classify S or noise labels as P phases). The F1-score weighs the precision and recall values equally and represents the harmonic mean of both values. F1-scores close to 1 suggest a good balance of high precision and recall for the model. The F1-score is given by 7

$$F1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (3)$$

4.3 Catalogue comparison

4.3.1 PNR-1z downhole data set

To complement the phase classification tests, we also compare the origin time catalogue generated from arrival picks of each model when applied to the PNR-1z continuous downhole data set. The initial (CMM) method catalogue serves as our benchmark. We analyse the proportion of events recalled from the CMM catalogue, new candidate seismic events (i.e. prior to quality control of the events) and the events missed by the DL models.

Our workflow is structured to group phase picks into events, associate phases (P to S) and then locate events to produce an origin time catalogue. We start by constructing an event catalogue

using phase picks output from each model across multiple geophones in the borehole array. Here, we define a candidate seismic event as a group of P phase picks from at least four different stations within a fixed 0.2 s time window. We choose a 0.2 s window because it corresponds to the latest traveltimes for a phase of the same event to travel from one end to the other end of the borehole array, given the available velocity model (doi: 10.5281/zenodo.13135600). We set the event defining threshold to at least four different stations because it is the typical minimum number of stations required to constrain location in time and space.

Next, we employ a straightforward approach for P to S phase association. We impose a fixed-time window from the initial P arrival time at each station. Specifically, we use a time difference of 0.3 s. This assumes that any S phase that is picked within the time range $t_p < t_s \leq t_p + 0.3$ will be associated with the initial P pick. The fixed time difference of 0.3 s theoretically corresponds to a 1.8 km radius. This time window for phase association adequately encompasses all the events in the target area (see Fig. S1, Supporting Information). Finally, we use the Non-Linear Location (NonLinLoc) algorithm (Lomax *et al.* 2000) to obtain origin times. For the full workflow, we conduct the model tests with one NVIDIA GeForce RTX 2080 Ti GPU on a cluster node with one CPU.

Each DL model has a user-defined detection threshold parameter. When the probability of a phase (P or S) crosses that threshold, the model picks and labels the arrival in time. As Zhu & Beroza (2019) recommended the lowest threshold (0.3) compared to the other models, we set the same probability threshold for most of the models (GPD, U-GPD and PhaseNet). However, initial detection tests indicated that EQT requires a lower threshold to detect events on the PNR-1z data set, so we use a threshold of 0.1 for EQT. Compared to the other models, EQT’s phase picking performance is significantly influenced by overlap (Pita-Sllim *et al.* 2023). We conducted a performance test for EQT using the default (0.3) overlap and optimized overlap (0.92) suggested by Pita-Sllim *et al.* (2023) and found that it took approximately 8.8 times longer with the 0.92

overlap (14 hr 36 min and 14.1 s) to run on an hour of test data. Based on these initial parametric tests, we select an overlap of 0.3 to achieve a balance between computational runtime and picking performance.

We compare the event origin times derived from DL model phase picks with the CMM catalogue. We assume that DL events within a window of ± 0.25 s of an origin time listed in the benchmark (CMM) catalogue are considered matches. We label any event that is not a match in the CMM catalogue as a new candidate event. In Fig. S2 (Supporting Information), we show that 99 per cent of the origin time residuals between the PhaseNet and CMM catalogues are within a ± 0.07 s window. The small origin time residuals validate the adequacy of the ± 0.25 -s window for matching DL phase pick-derived event origin times with the CMM-catalogued events.

4.3.2 Selected time-series

To provide a more detailed comparison, we focus on two different periods of interest to assess each model: on 2018 December 11, from 9am to 10am, and from 11am to 12pm. We select an hour of data when the injection rate and seismicity were high (9am to 10am, Fig. 2b) and an hour when the seismicity was relatively quiet and injection was paused (11am to 12pm, Fig. 2c). We can thus observe which models are able to pick phases during a very active period (i.e. with high event rates) and during a quieter period of seismicity. The 11am to 12pm period also contains the largest (M_w 1.1) earthquake in the data set. We additionally use this period to assess whether the models detect more aftershocks after the M_w 1.1 event.

4.4 Candidate seismic event validation using LinMEF: a Linear Moveout Event Filter

As DL models have demonstrated the ability to detect a substantial number of new events (Beroza *et al.* 2021; García *et al.* 2022), we develop an automated validation method to ensure that the new events originate from our target area, i.e., from within the zone of the hydraulic fracturing stimulations. We visually evaluate the subsets of the new events to ensure that we do not exclude potentially interesting events (see Section 5.2).

Our method exploits the expected moveout of phase arrivals across a near-linear borehole array (Fig. 3). By imposing the near-linear moveout pattern, we filter the model detections (candidate events) to focus on events arriving from the PNR-1z well stimulations. This method leverages the information gathered from multiple geophones, taking advantage of the wealth of data obtained from a borehole array of individual geophone observations. We restrict our approach to just the P picks of each new event, as the DL models picked more P phases compared to S phases.

Because the hydraulic fracturing stimulations occur below the borehole array, we make two key assumptions: first, the P picks should exhibit moveout from the deepest to the most shallow geophone, and secondly, the P -wave velocity across the array should approximately be the velocity of a P -wave travelling through shale (4700 m s^{-1}). These assumptions allow us to filter candidate events that deviate from the expected near-linear moveout, thereby retaining only the seismicity originating from the stimulated volume of rock (i.e. from the in-zone).

To validate the thousands of new machine learning-picked detections, we utilize the above information and apply a method we

call the Linear Moveout Event Filter (LinMEF) to the seismic phase picks of each event. LinMEF is a simple and robust method consisting of three steps (Fig. 3).

First, for each new event, we use the group of P picks across the borehole array associated with the specific event origin time. We obtain a list of time differences t_{diff} between each pick time in the group, t_{pick} , and the earliest arrival time, t_0 :

$$t_{\text{diff}} = t_{\text{pick}} - t_0. \quad (4)$$

When plotting each t_{diff} against each corresponding station depth, we observe that the data points roughly follow a straight line for events arriving from the in-zone (Fig. 3b). The inverse of this gradient is representative of the P -wave velocity.

Secondly, after calculating t_{diff} from all available picks across the array for all events, we choose a random subset of 250 newly identified candidate events as our calibration data set. We visually check the pick moveouts for each event and labelled this calibration data set as either ‘target’ or ‘non-target’ events depending on their moveout (Fig. 3c). ‘Target’ events refer to events that: have a near-linear moveout; arrive from the bottom to the top of the array; and have apparent velocities close to the P -wave velocity in shale (4700 m s^{-1}) shown in Fig. 3(a).

‘Non-target’ events are candidate events that characteristically deviate from the ‘target’ events. We further discuss these non-target events in Section 5.2.

We use this data set to constrain the range of gradients (i.e. the approximate inverse of P -wave velocity) of the P picks that belong to ‘target’ and ‘non-target’ labelled events. We estimate the L1-norm best-fitting line of the arrival pick times across all stations for every event. The L1-norm reduces undesired effects from pick time outliers on the arrival time gradient estimates, as compared to the L2-norm. Fig. 3(c) shows that the linear regression slopes of ‘target’ and ‘non-target’ labelled events fall into a bimodal distribution.

Lastly, we calculate the 90 per cent range (using the 5th and 95th percentiles) of the resulting gradients around the ‘target events’ labelled distribution to define our event filter. We subsequently classify any new candidate events with a linear moveout slope within this 90 per cent range as ‘target’ events and any that fall outside this range as ‘non-target’ events.

5 RESULTS

5.1 Picking assessment: classification test results

We compute precision, recall and F1-scores from the confusion matrices of each model (see Supporting Information). The classification test results show that PhaseNet outperforms other models in terms of precision and recall (Fig. 4). EQT follows closely with high precision and recall values (0.81 and 0.77, respectively). The U-GPD and GPD models have precision and recall values ranging between 0.44 and 0.64, implying occasional misclassifications. Overall, PhaseNet consistently demonstrates the highest precision, recall and F1-score values, highlighting its accuracy in classifying known seismic phases. Most models exhibited superior performance in classifying individual P phases compared to S phases (Fig. 4f). Higher F1-scores for P -phase classifications also suggest that the DL models more successfully identified P phases than S .

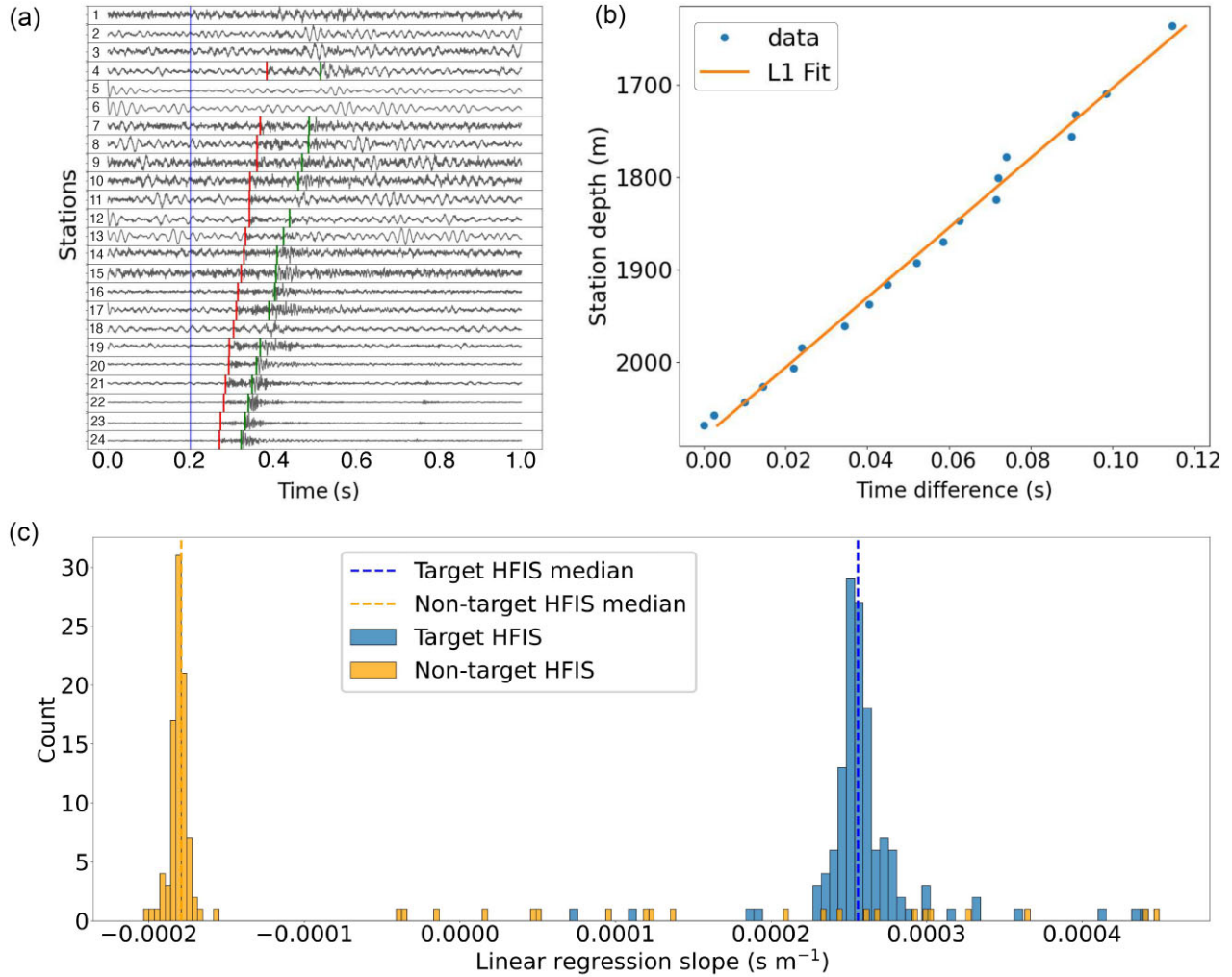


Figure 3. (a) An example of a new event picked by PhaseNet with P (red) and S (green) arrival picks arriving across the borehole array (geophones numbered on the side of each seismogram from 1 to 24). The vertical blue line represents the event origin time. (b) Time differences between the earliest P arrival pick time (t_0) and each individual pick time (t_{pick}) of the new event against station depth. We fit an L1-norm best-fitting line to the data points. (c) A histogram of estimated linear regression slopes (slowness) of 250 random new candidate seismic events. The target events (blue) and non-target events (orange) labelled from manual visual inspection. The dotted lines represent the median value of the gradients for each label (target and non-target HFIS). Panels (b) and (c) illustrate the process of candidate event validation using the LinMEF.

Visual inspection of PhaseNet’s phase misclassifications revealed heightened noise levels around the phase in the data or when a larger event was present within the normalized window, making the smaller magnitude target phase located in the middle of the window appear smaller and more challenging to detect.

In terms of the picking precision of each model, PhaseNet consistently demonstrates the most precise and accurate picks (Fig. 5). In contrast, the GPD model exhibits less precise picking due to its output of a single phase probability value for each data window shift, leading to ambiguity in the arrival timing. The PhaseNet and U-GPD models, which follow a U-Net styled architecture, show precise picking with temporally continuous probability traces. Fig. 5 illustrates the phase arrival picks from each model, clearly indicating the precise phase picking by the U-GPD and PhaseNet models, while the GPD model produces imprecise picks. It should be noted that the EQT model, although capable of picking phases precisely, frequently misses phase arrivals in general.

5.2 Characterization of non-target events

We observe that the DL models consistently detect events or phase arrivals that we do not believe to be induced seismic events associated with the injection. Figs 5 and 6 illustrate the two types of candidate events detected, namely ‘target events’ and ‘non-target events’, respectively. The non-target events belong to at least four types of events that do not exhibit characteristics indicative of induced seismicity resulting from the hydraulic fracturing stimulations through the PNR-1z borehole. We label these signals as ‘zig-zag’ events, ‘tube waves 1’, ‘tube waves 2’ and ‘emergent’ events. Through visual observations and event characterization based on apparent velocities and angle of incidence, we determine that these signals are likely unrelated to our target events.

‘Zig-zag’ events are waves that arrive from the opposite direction (from shallow to deep) compared to induced seismicity (from deep to shallow). They have an average apparent velocity of 7500 m s⁻¹. This is significantly faster than the P -wave velocity in shale (~ 4700 m s⁻¹) and in the steel casing (~ 4500 m s⁻¹). Most

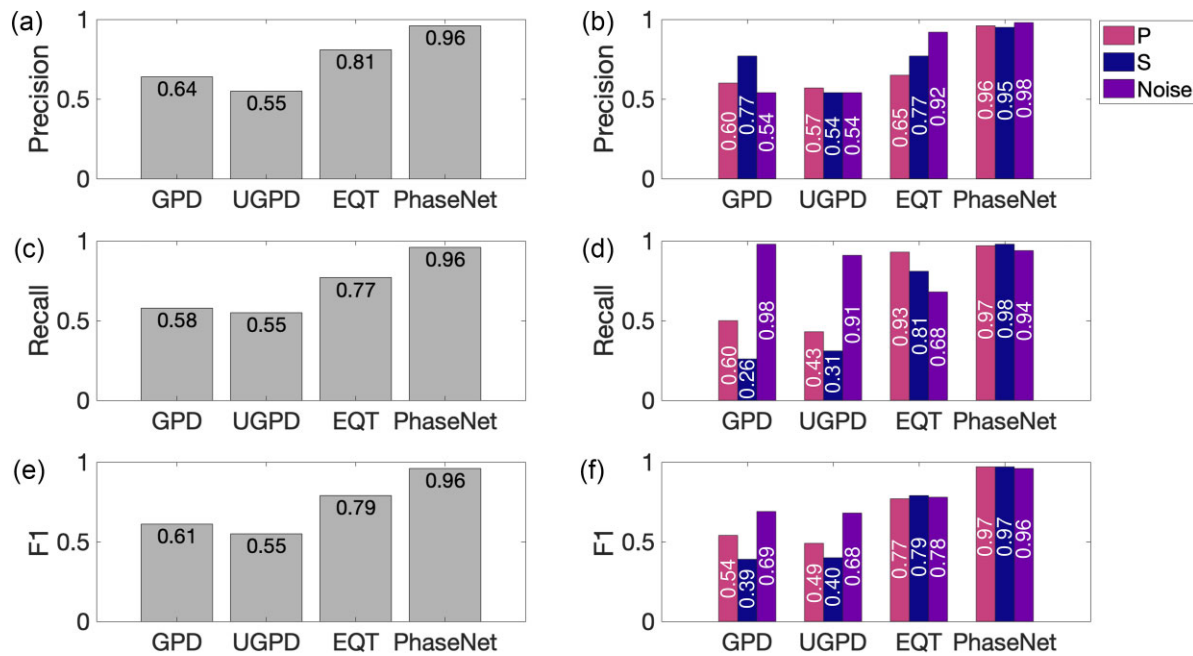


Figure 4. The left-hand column of panels shows the (a) overall precision, (c) overall recall and (e) overall F1-scores of each model. The right-hand column shows the individual *P* (pink, left bar), *S* (dark blue, middle bar) and noise (purple, right bar) phases of the (b) precision, (d) recall and (f) F1-scores.

of these events do not have *S* picks, and they only occur during working hours, between 8am to 4pm.

‘Tube waves 1’ are low frequency and travel at an average apparent velocity of 1445 m s^{-1} , which is slower than the *P*-wave velocity in shale but close to the velocity of *P* waves in water ($\sim 1500 \text{ m s}^{-1}$). Tube waves are a product of Rayleigh waves interacting with the wellbore, propagating along the walls of a fluid-filled borehole (Sheriff 2002; Gadallah & Fisher 2008).

‘Tube waves 2’ are higher frequency tube waves that appear to arrive at the broadside of the borehole array and travel from between the geophones on the PNR-2 well throughout the array. They do not arrive at any particular single station and their average velocity is similar to that of tube waves ($\sim 1445 \text{ m s}^{-1}$).

The ‘emergent’ events, which occur at high frequencies and do not follow a consistent moveout, present a unique challenge. We observe a gradual energy buildup in these arrivals, rendering it challenging for the models to precisely identify the phase onset. The emergent nature of these events complicates picking, which results in an inconsistent moveout pattern. These events arrive almost instantaneously, though the arrival patterns vary randomly along the array. In addition, both the frequency content and amplitude of these events deviate characteristically among the geophones. None of these events correspond to any catalogued regional earthquakes in UK. Given the unusual observations, we conclude that these ‘emergent’ events are not HFIS.

We visually label a random sample of 250 candidate seismic events detected by the best-performing DL model, PhaseNet. Our observations show that 55.6 per cent (139) of these events were induced seismicity with a near-linear *P*-wave moveout (i.e. target events), while 34.8 per cent (87) were ‘zig-zag’, 2.8 per cent (10) corresponded to ‘tube waves 1’ or ‘tube waves 2’, 2.4 per cent (11) were ‘emergent’ events and 1.2 per cent (3) were false positives. When we manually assess the residual percentage of non-HFIS events, we find that LinMEF successfully eliminates 93.7 per cent (104) non-target events from the test set. LinMEF decreases the non-target event rate from 44.4 per cent to 2.8 per cent. To verify

the detection of HFIS using these DL models, we implement the LinMEF method (detailed in Section 4.4) to robustly eliminate these non-target events from the DL event catalogues.

5.3 Model catalogue comparison

Fig. 7 shows that PhaseNet successfully recalls the most previously catalogued events (36 75 events, 94.6 per cent) from the full continuous data set, missing only 5.4 per cent (2077 events) of the CMM events. Additionally, PhaseNet detects the highest number of undetected events (+41.2 per cent), corresponding to 15 35 new events after filtering out non-target events using LinMEF. Comparatively, U-GPD outperforms GPD as it recalls more catalogued events (76.6 and 59.5 per cent, respectively) and detects more new events (8302 events, +21.6 per cent and 1918 events, +4.99 per cent, respectively). On the other hand, EQT struggles to detect events, identifying around 45.8 per cent of the CMM catalogue (17 11 events) and 809 new events (+2.1 per cent). We show the event locations of these catalogues in Fig. S3 (Supporting Information).

Most of the models begin to miss events at magnitudes below $M_w -0.5$ (Fig. 8). Additionally, we note the magnitude ranges for the recalled and missed events of each model. Fig. 8 shows that EQT started missing events with M_w below 0.7, suggesting its limitation in recognizing microseismic phases within the higher frequency data set. PhaseNet recalls events from benchmark (CMM catalogue) very well (over 90 per cent) over the $-3 \geq M_w \geq 1$ range. We also show that U-GPD can recall events down to lower magnitudes more so than the GPD model.

We remark that most DL models’ phase picks are not affected by clipping by visually checking the picks of 100 of the largest events with clipped amplitudes. The modelled *P* and *S* phase picks for GPD, U-GPD and PhaseNet are still picked accurately. However, the detection performance of EQT might be affected by amplitude clipping. Although EQT detects both *P* and *S* phases of the largest clipped event, the earthquake detection probability trace shows a steep drop off where the *S* wave is clipped and lower *S* arrival

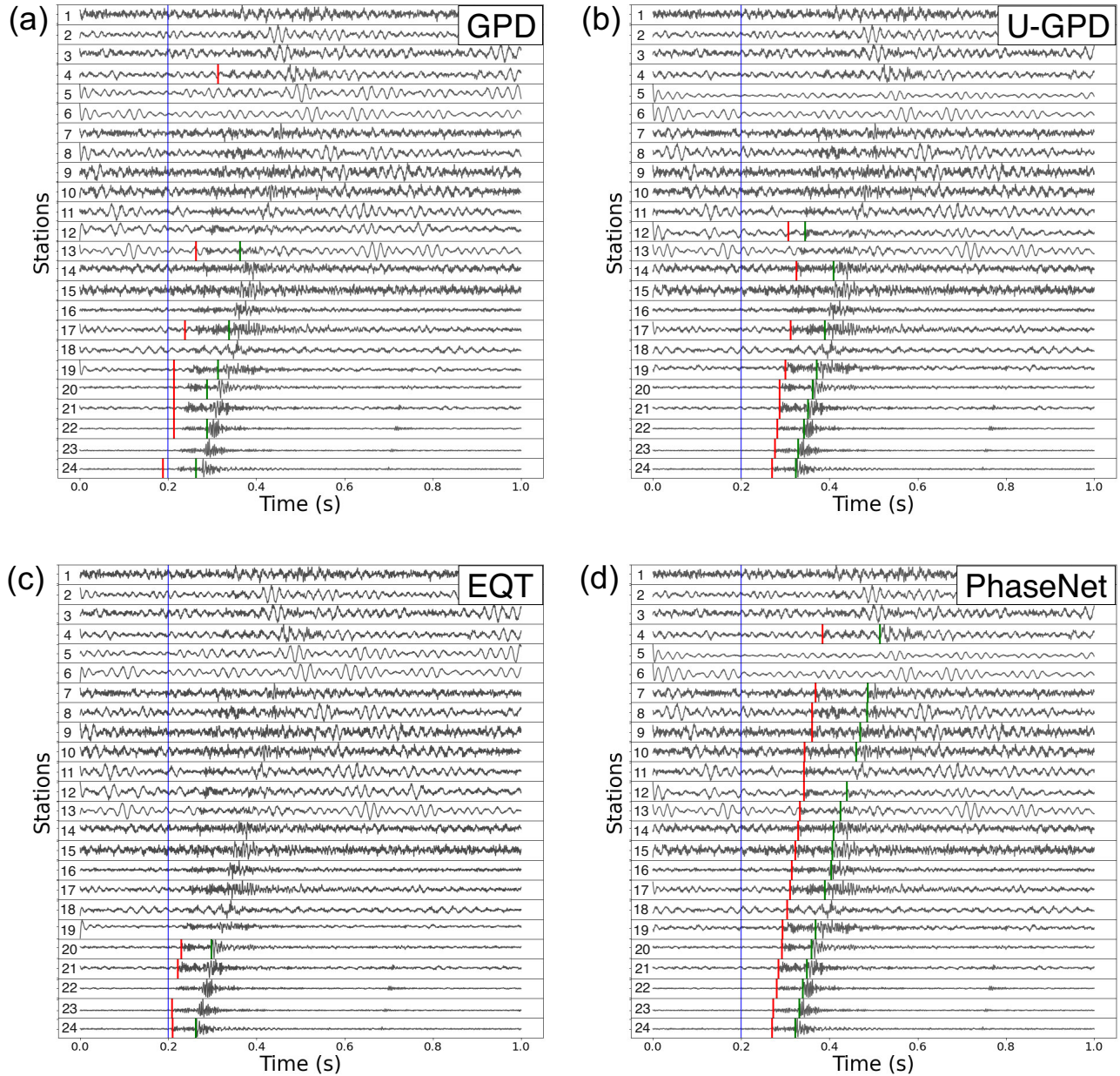


Figure 5. Seismograms (Z component) of the same event across the borehole array from Stations 1 (top row, most shallow) to 24 (bottom row, deepest) showing the phase classification and picking precision with P picks (red vertical lines) and S picks (green vertical lines) from the (a) GPD, (b) U-GPD, (c) EQTransformer and (d) PhaseNet models. The blue vertical line across all stations is the inferred event origin time.

probability values (Fig. S4, Supporting Information) compared to unclipped events (Fig. S5, Supporting Information). These observations explain the unstable decrease in recall rates for the clipped range of magnitudes ($M_w > -0.3$) in Fig. 8 followed by a more stable recall rate decrease for events $M_w < -0.5$.

Results from the selected hours show that during the injection period (Fig. 9), both PhaseNet and U-GPD successfully recall a majority (>90 per cent) of the CMM catalogue, while also detecting the highest number of new events within the hour. During high injection rates, U-GPD detects more new events than PhaseNet, whereas the reverse is true during the quieter period. We also observe increases in seismicity rates during both selected hours for PhaseNet and U-GPD compared to the CMM method.

Figs 10(b) and (d) demonstrate that PhaseNet outperforms U-GPD during the quieter period, with the exception of struggling to detect small events after the initial largest event when a cluster of aftershocks occurred. This suggests that PhaseNet's performance is adversely affected by high seismicity rates. Furthermore, we note that PhaseNet begins missing more small events during high seismic activity, likely due to the smaller amplitudes of these events as normalization may be biased towards the increased presence of larger events.

5.4 Model runtimes and computational efficiency

We present the model runtime statistics and performance metrics based on one hour of continuous borehole data (Table 2). We use the

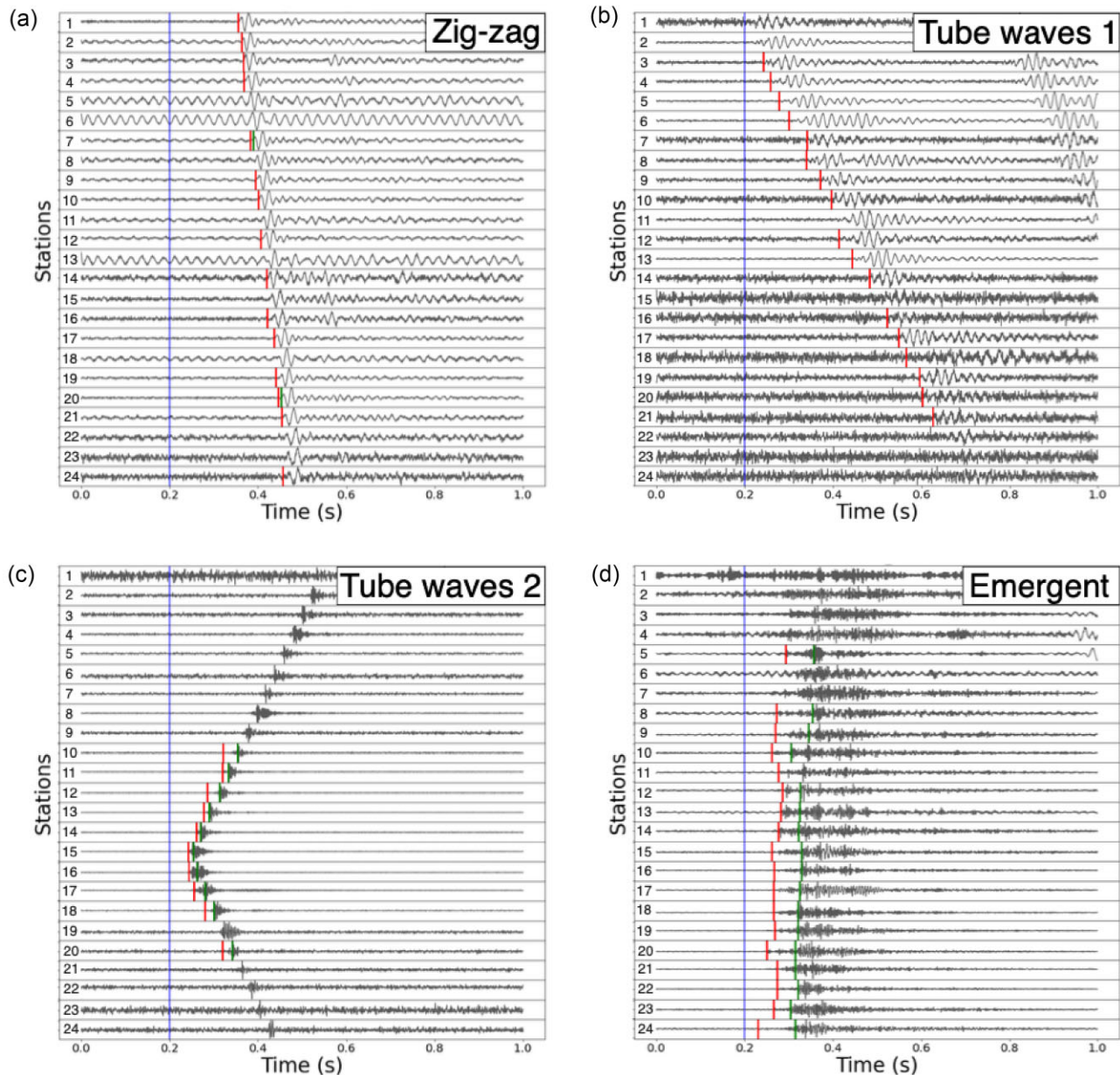


Figure 6. Seismograms of the four types of non-target events: (a) ‘zig-zag’, (b) ‘tube waves 1’, (c) ‘tube waves 2’ and (d) ‘emergent’ events across the borehole array from Stations 1 (top row, most shallow) to 24 (bottom row, deepest) showing the *P* picks (red vertical lines) and *S* picks (green vertical lines) from PhaseNet. The blue vertical line across all stations is the inferred event origin time.

NVIDIA Tesla K80 12 GB GPU on Google Colab for these tests. Our findings reveal that that most DL models (with the exception of EQT) take less time to process than the given data window length (one hour of data). In ascending order, PhaseNet, GPD and U-GPD demonstrate processing speeds surpassing real time by 3.72, 4.94 and 12.75 times faster, respectively. Although these figures might seem modest, it is crucial to consider the high sampling frequency of our test data (2000 Hz). One hour of 2000 Hz data are equivalent to 20 hr of 100 Hz data. When scaled to standard 100 Hz data, the processing speed of the models are 74.4, 98.8 and 255 times faster, respectively. These results highlight the remarkable efficiency of these models, highlighting their suitability to real-time microseismic monitoring.

6 DISCUSSION

The results demonstrate the applicability of U-GPD and PhaseNet to high-frequency borehole data, although to varying degrees. Our study shows that, in particular, PhaseNet outperforms the CMM method, as it recalls up to 95 per cent of the catalogue and detected approximately 15 800 new events. In total, PhaseNet detects over 52 000 target events. Although U-GPD detects fewer earthquakes than the CMM method, it still identifies a significant number of events within the data set (over 37 800). GPD and EQT (detected over 17 800 and over 24 900, respectively) can detect events in high-frequency borehole data, but EQT yields fewer results due to its picking performance, while GPD’s limitations are due to both detection performance and picking precision. Most models (GPD,

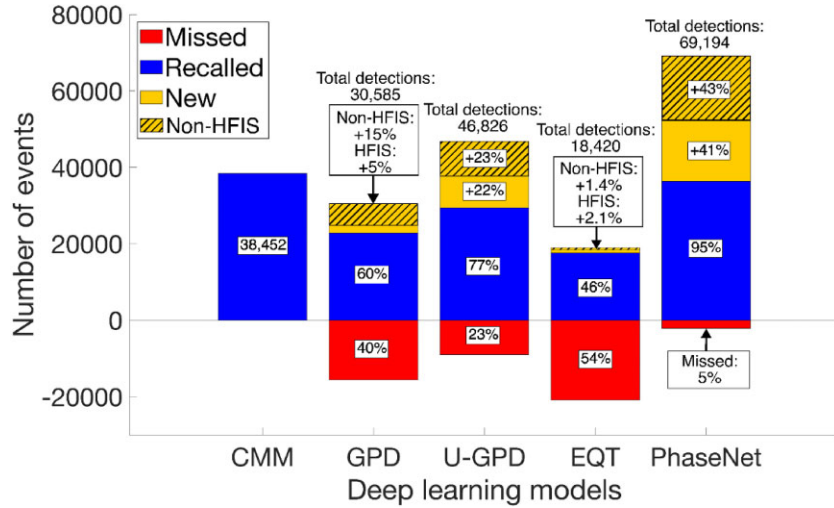


Figure 7. A comparison of the total number of model event detections in the full continuous PNR-1z data set, including the new (yellow), recalled (blue) and missed (red) events. We filtered the new candidate seismic events using the LinMEF method, so the cross-hatched yellow sections represent the number of non-target events, and the non-hatched yellow sections illustrate the number of target events (i.e. QC'd new events). The annotations in squares refer to the percentage with respect to the total number of CMM-detected events.

U-GPD and PhaseNet) had no issue picking phases on amplitude-clipped events.

Although EQT displays good classification scores (precision of 0.81 and recall of 0.77), it underperforms in consistently detecting events with M_w below 0.7 in high-frequency borehole data. As shown in Fig. 8, and Figs S4 and S5 (Supporting Information), EQT struggles with detection on clipped data ($M_w > -0.3$), leading to inconsistent recall rates. However, this does not completely account for the unclipped missed events ($M_w < -0.3$). These smaller missed events could mean that EQT is also encountering challenges with its attention mechanism as it does not perform as well in 2000 Hz data compared to 100 Hz data. This issue arises from the deeper, more complex model learning data representations that may not generalize well to higher frequency data. The attention mechanism in EQT, which predicts the arrival of seismic phases after determining the presence of an earthquake, may face difficulties when the P and S phases arrive at different positions and appear differently in the 2000 Hz data compared to the 100 Hz training data. Consequently, the complex transformer-based EQT model struggles to detect lower magnitude events with low SNR in high-frequency data.

The precision of phase picking heavily depends on the network architecture. Models like U-GPD and PhaseNet, which utilize fully convolutional networks, provide more precise picks by leveraging higher resolution continuous probability phase traces. Similarly, EQT leverages these continuous probability traces to achieve precise phase picks. The pick precision of the GPD model is influenced by its combination of convolutional and FCNNs, which output a single probability value per class for each data-window shift. Therefore, in GPD's case, pick precision is significantly dependent on the size of the window shift across the data.

PhaseNet outperforms other models in terms of precision, recall and F1-score for microseismic phase classification. It exhibits high metric scores for both P and S arrivals, while other models show inconsistencies in classifying P and S phases. GPD, U-GPD and EQT have better metric scores for classifying P phases compared to S phases due to S waves characteristically having lower SNR.

PhaseNet's success on this data set may be attributed to its exposure to different types of instrument data during training. The

model's training incorporated seismic data from several frequency bands and instruments (Zhu & Beroza 2019), resulting in an apparently improved generalization for phase picking in the HFIS setting. Furthermore, the relatively small model size of PhaseNet (10 layers, 269 675 parameters) combined with a large training data set reduces the risk of overfitting to its initial training set (623 054 training samples). Although U-GPD has a similar architecture and model size to PhaseNet (9 layers, 672 419 parameters), it displays signs of overfitting to the small volcanic data set (in 100 Hz) used to fine-tune the model (Lapins *et al.* 2021). Overfitting can be mitigated through stronger data augmentation, such as resampling the training data to lower and higher sample rates to vary the number of samples between a P and S phase and expanding the training data set to include more diverse events with different P to S time differences. Lastly, as the deepest of all models tested (56 layers, 376 935 parameters), EQT might be susceptible to overfitting because of its number of layers and the lack of data diversity of its original training set (1.3 million samples). The deep transformer-based model might have learned more complex and abstract features that is more tailored to the 100 Hz training data as the number of layers increase (Goodfellow *et al.* 2016) and therefore, EQT might struggle to recognize extracted features in higher frequency data.

As observed in Fig. 10, U-GPD performs better at picking events during high seismic rates due to its smaller window size (400 samples) compared to PhaseNet (3000 samples). This enables U-GPD to handle the presence of multiple events within the same temporal data window more effectively. Although PhaseNet struggles (relative to U-GPD) during higher event rates, it still detects more small events ($M_w \leq -2$) than U-GPD overall (Fig. 10).

The limitations of the CMM method become apparent when multiple events overlap within the same user-defined time window (see Fig. S6, Supporting Information). When there is more than one event within this window, CMM detects only the largest event based on energy maxima (Drew *et al.* 2013), potentially missing smaller events. Reducing the time step window in the CMM method could enhance its performance, but this would increase its computational intensity and overall cost, thereby limiting its feasibility.

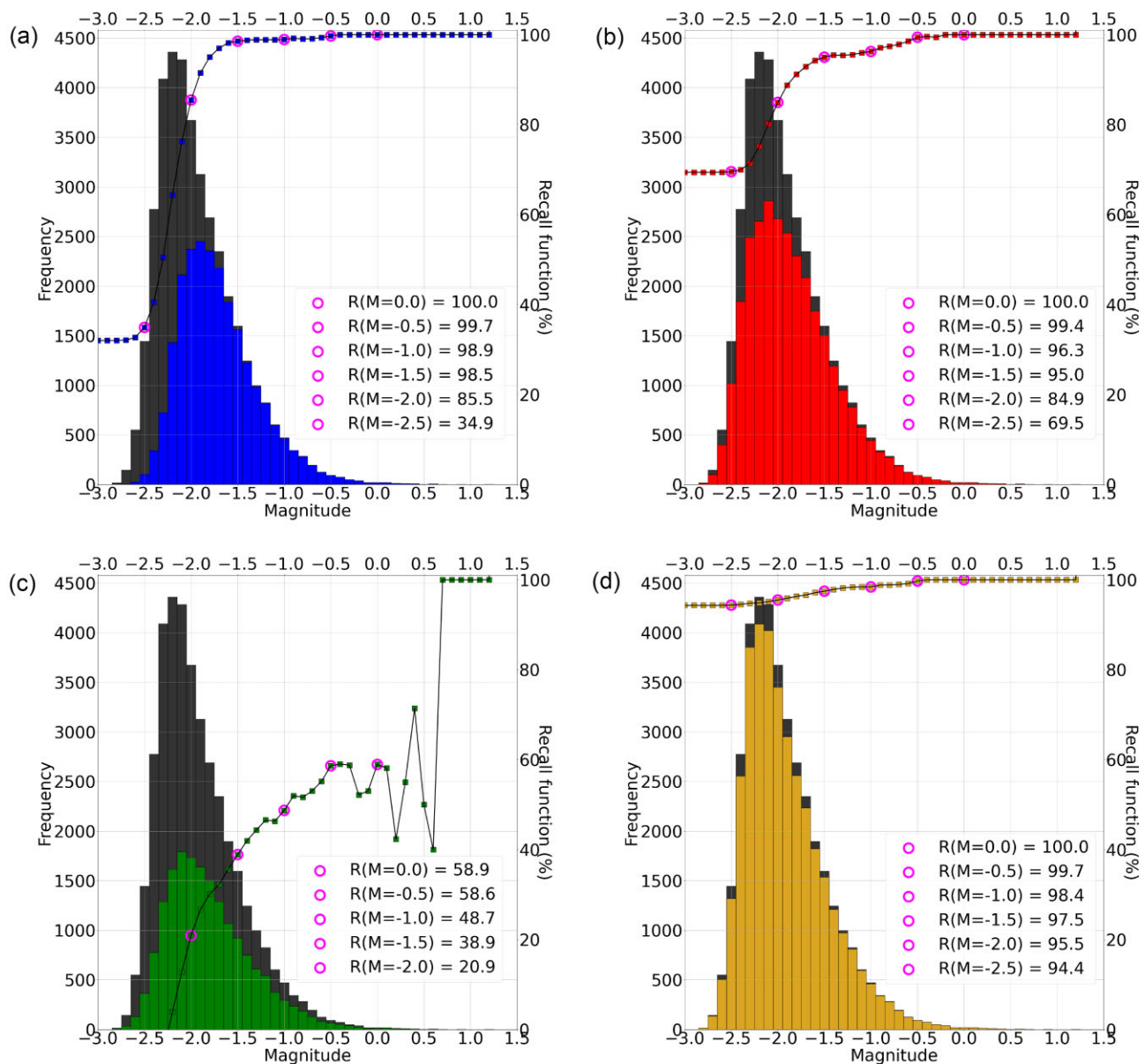


Figure 8. Magnitude distributions for the recalled events detected by (a) GPD (blue), (b) U-GPD (red), (c) EQT (green) and (d) PhaseNet (yellow) compared to the benchmark CMM (black bars) catalogue. The recall function (black line) of each model catalogue is plotted with their respective marker colours. The recall function shows the percentage of CMM events recalled from the benchmark for a given magnitude bin, M . Magenta circles in the legend mark the recall for magnitude bins at M_w 0, -0.5 , -1 , -1.5 , -2 and -2.5 . We use the estimated magnitudes from the CMM catalogue so new events are not included in this plot.

In contrast, DL models have the potential to pick out overlapping events (Fig. S6, Supporting Information). However, our current approach to phase association and event grouping lacks the sophistication needed to differentiate multiple closely arriving events within a fixed time window. Future research should consider more advanced phase associators such as REAL (Zhang *et al.* 2019), an optimized grid-search algorithm, GaMMA (Zhu *et al.* 2022), which considers arrival time moveout and amplitude decay with distance, or PyOcto (Münchmeyer 2023), a 4-D space-time partitioning algorithm to separate overlapping events in time. These more advanced techniques could leverage DL-detected phases arriving in close temporal proximity, thereby significantly enhancing event detection.

All models might benefit from additional training or exposure through fine-tuning with microseismic data sets ($-3 \leq M_w \leq 0$)

to improve their ability to recognize smaller events recorded at higher sampling rates. However, PhaseNet already exhibits promising performance and can be deployed without any modification. The events missed by PhaseNet are primarily small events ($M_w \leq -0.5$) that occur during high seismic rates, where larger events may overshadow the smaller earthquakes. Shortening PhaseNet's input window length could potentially isolate smaller events for detection, but there would be a trade-off with increased computational time due to the greater number of windows to be processed. Since most of these models can be employed in near real-time, finding a balance between efficiency and effectiveness is crucial. The efficiency of DL models makes them suitable for automated detection methods in real-time monitoring.

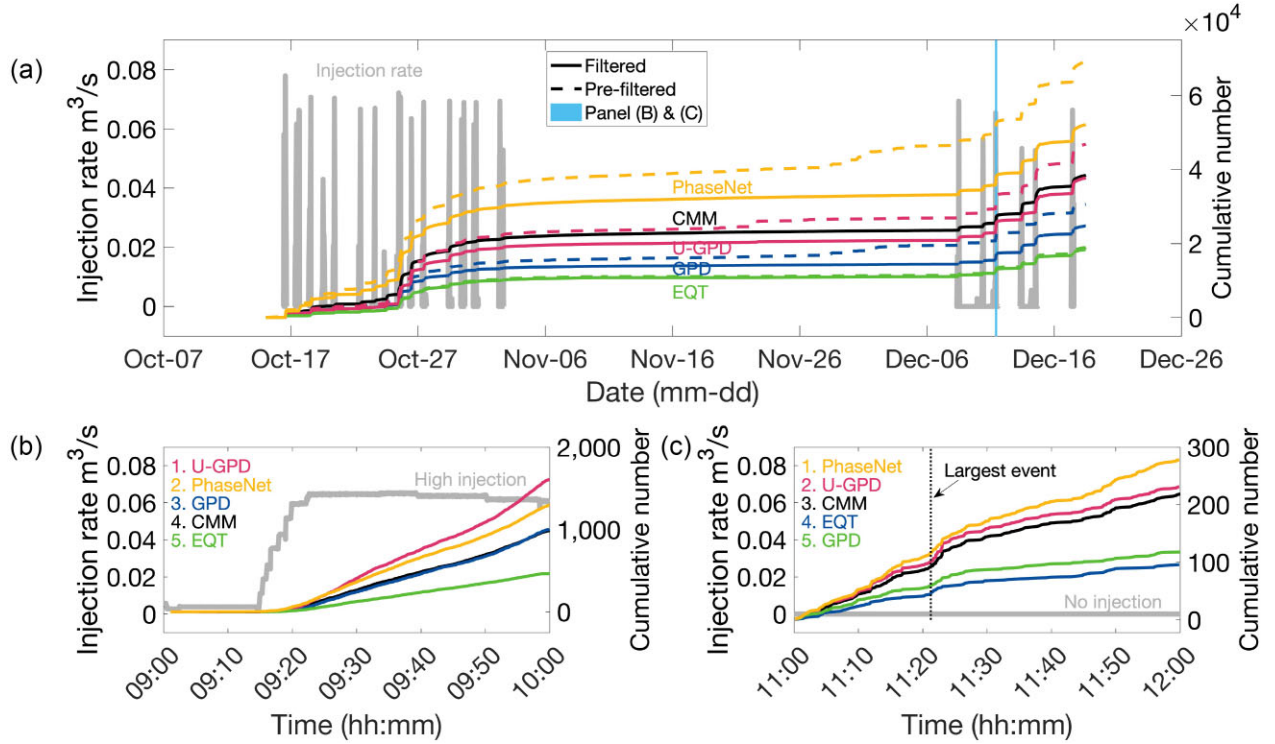


Figure 9. Temporal plots of the cumulative number of model detections (right y-axis) (a) across the PNR-1z continuous data set with injection rates overlaid (in grey, left y-axis) for PhaseNet (orange), U-GPD (red), CMM (black), GPD (blue) and EQT (green). The solid lines represent the cumulative number of events in the LinMEF-filtered catalogue, whereas the dashed lines show the cumulative number of the pre-filtered catalogue. Cumulative number for each model (b) during high injection rate and high event rate, and (c) during no injection and when the largest PNR-1z event occurred. The inset list of models in (b) and (c) represent the performance ranking of the DL models and CMM method.

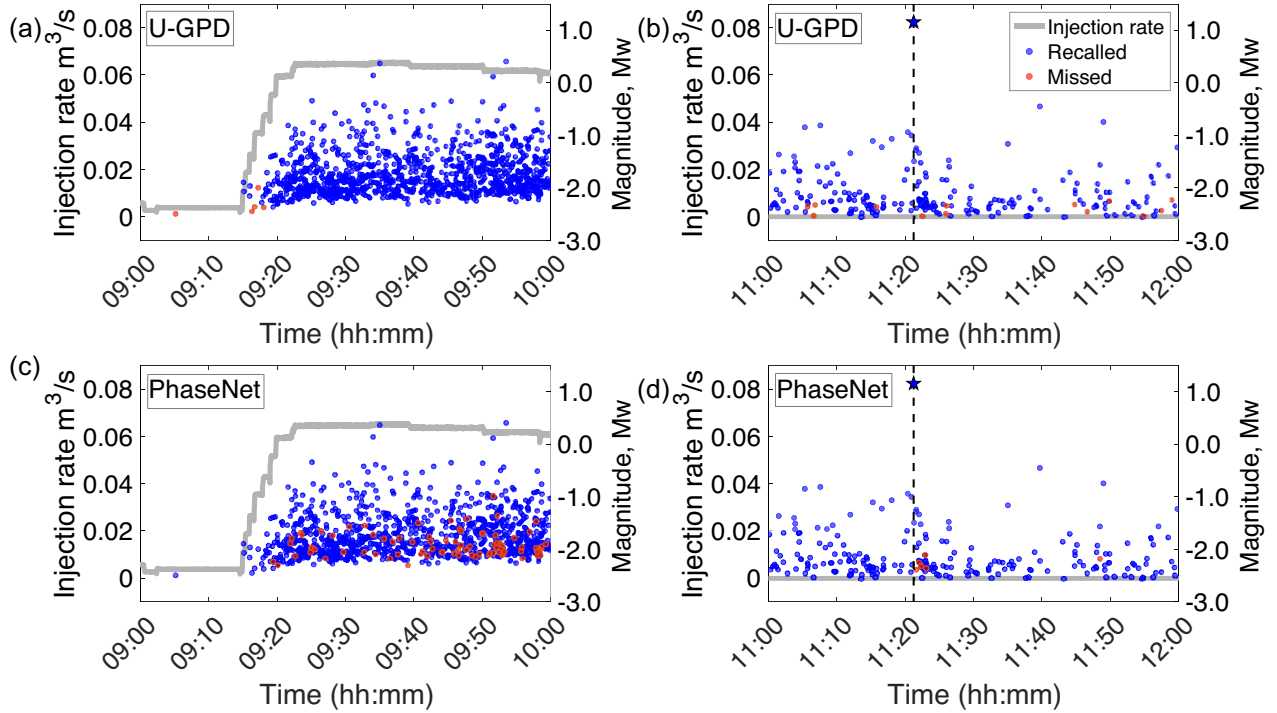


Figure 10. (a) and (c) Time-magnitude plots of the high injection rate hour and (b) and (d) the quieter period with injection rates (black line, left-hand axis) for the U-GPD (top row) and PhaseNet (bottom row) models. Missed (red) and recalled (blue) events are plotted with their respective moment magnitudes, M_w (right-hand axis). The largest earthquake ($M_w 1.1$) is plotted as a star and is denoted with a dashed line.

Table 2. Model runtime statistics for GPD, U-GPD, EQTransformer and PhaseNet on Google Colab, using an NVIDIA Tesla K80 12 GB GPU. These processing times are for one hour of PNR-1z data (2000 Hz), which is equivalent to 20 hr of 100 Hz data. We also include results (recall rate and number of new events) for the selected hour during high injection.

Models	Time taken (24 stations)	$\frac{\text{Data window length}}{\text{Time taken}}$	Recall rate (n events)	Number of new events
GPD	12 min 8.53 s	4.94	87.5 per cent (863)	143 (+14.5 per cent)
U-GPD	4 min 42.37 s	12.75	99.5 per cent (981)	632 (+64.1 per cent)
EQT	99 min 35.7 s	0.60	68.2 per cent (672)	331 (+33.6 per cent)
PhaseNet	16 min 5.88 s	3.72	91.6 per cent (903)	396 (+40.2 per cent)

7 CONCLUSIONS

In this study, we compare the ability of four DL models (GPD, U-GPD, EQT and PhaseNet) to detect HFIS through phase picking in high-frequency (2000 Hz) downhole data. We find that PhaseNet is most suitable for microseismic monitoring applications, when comparing the phase classification abilities of the models and their resulting event detections. PhaseNet shows the best phase classification results and the most precise picking compared to the other models.

Based on our evaluation, we conclude that while additional transfer learning might further enhance the performance of PhaseNet, it can still be readily applied ‘off-the-shelf’ (recall rate of 94.6 per cent) for downhole monitoring of induced seismicity with moment magnitudes M_w greater than -0.5 . Models such as GPD and U-GPD (recall rate of 59.5 and 76.6 per cent respectfully) require transfer learning to improve their microseismic detection, to detect events with M_w below -0.5 . EQT, on the other hand, currently requires retraining to detect both clipped events as well as events below M_w 0.7 for it to be effectively applied to high-frequency downhole data.

We infer that model architecture and exposure to different training data influences microseismic phase detection and picking precision, which are important for event location and moment magnitude estimation. Network architectures such as fully convolutional U-Nets with a small model size that provide continuous high-resolution probability traces offer more precise phase picking and better generalization to pick events in high-frequency data compared to more complex transformer models (e.g. EQT) or models comprising fully connected layers (e.g. GPD).

In addition to the over 38 000 catalogued earthquakes already detected on site, PhaseNet identified around 15 800 new events occurring within the PNR-1z data set. The identification of thousands of additional events could lead to improved observations of the spatiotemporal characteristics of HFIS within the context of various driving mechanisms (e.g. Kettlety & Verdon 2021; Herrmann *et al.* 2022) and facilitate the real-time tracking of seismicity during well stimulations.

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SUPPORTING INFORMATION

Supplementary data are available at *GJIRAS* online.

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DATA AVAILABILITY

All seismic data and microseismic catalogue for PNR-1z operations are publicly available through the North Sea Transition Authority (<https://www.nstauthority.co.uk/exploration-production/onshore/onshore-reports-and-data/preston-new-road-pnr-1z-hydraulic-fracturing-operations-data/>). All code and data to run the classification tests are available at <https://doi.org/10.5281/zenodo.13135600>. The model earthquake catalogues for PNR-1z and velocity models are also available in the same repository.

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